

Jimin Han

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OBJECTIVE

Undergraduate mechanical engineering student seeking opportunities to apply CAD, prototyping, and computational modeling skills to interdisciplinary research and engineering projects. I aim to deepen my experience with simulation and experimentation while contributing to solutions that improve real-world systems and user experiences.

EDUCATION

Rice University, Texas, USA

Aug 2025 – Present

B.S. in Mechanical Engineering and B.A. in Mathematics (expected May 2029), GPA: 4.0/4.0

SKILLS

Programming & Data: MATLAB, Python (pandas, NumPy, scikit-learn, XGBoost, SHAP, Matplotlib), LabVIEW;

Design & Analysis: SolidWorks, Onshape, QGIS, OpenRocket, DIC;

Manufacturing: 3D printing, waterjetting, laser cutting, machining, soldering;

Languages: Korean (native), English (fluent), Mandarin (working proficiency)

WORK EXPERIENCE

Tribomechadynamics Lab – *Undergraduate Research Assistant* – Rice University

Oct 2025 – Present

Research Advisor: Prof. Matthew Brake

- Compiled a 40-item, \$1.5K bill of materials for a mini split-Hopkinson pressure bar built on 4 mm Ti-6Al-4V bars to study high-strain-rate failure in bolted metallic joints
- Analyze DIC and strain gauge data in LabVIEW to extract stress-strain response from high-strain-rate tests
- Model bar components, test fixtures, and a T-slot and acrylic safety enclosure in SolidWorks; sourced parts across vendors

NSF RIDE REU (1 of 10 selected) – *Research Intern* – University of Massachusetts Amherst

Jun 2026 – Aug 2026

Research Advisor: Prof. Eric Gonzales

- Cleaned and combined 120,000+ GPS and battery records from an electric bus in Kigali, Rwanda with passenger counts from CCTV footage, elevation, and weather data
- Compared a physics-based model against three machine learning models; found that random train/test splitting leaked data and overstated accuracy, and switched to a 60-second block split that reversed which model performed best
- Presented findings in a research poster and first-authored a manuscript submitted to the Transportation Research Board

NSF ExLENT (1 of 20 selected) – *Research Trainee* – Michigan Technological University

Feb 2026 – May 2026

- Completed six-module mechatronics curriculum in circuit controls, hydraulics, robotics, PLCs, cybersecurity, and AI
- Applied the curriculum in person at Michigan Tech: programmed FANUC robots (frames, end-of-arm tooling, motion programs), wrote Allen-Bradley ladder logic with timers and counters, and built hydraulic circuits
- Visited four regional manufacturers, including PCB fabrication at Calumet Electronics and vacuum-chamber testing of electric propulsion at Orbion Space Technology

Lukkey AG – *Engineering Support Intern* – Remote

Dec 2025 – Jan 2026

- Evaluated a Swiss hardware Bitcoin wallet for Korean market fit against demand, regulation, and competitors
- Tested an early unit and reported usability and documentation feedback to the engineering team ahead of launch
- Localized the full technical terminology set into Korean and met with the team across 10 sessions to align on terms, KRW pricing, and deployment strategy

Xcellent Life – *Strategy Intern* – Nashville, TN

Jun 2025 – Jul 2025

- Analyzed regulatory, cultural, and logistical constraints across 3 healthcare-adjacent markets to guide market entry strategy with the CEO
- Built email automation templates that cut manual outreach time and drove investor contact with 10+ firms, leading to a multi-investor meeting

PROJECTS

Battle Bots – *Leader & Engineer at Rice Robotics Club* – Rice University

Sep 2025 – Present

- Designed and built 3 iterations of an antweight combat robot in Onshape (3D printing, waterjet, machining, soldering)
- Redesigned the weapon after its edge chipped in matches: blunted it, widened the striking face to 0.75", opened the shaft bore 0.100" for clearance, added a key, and fixed 0.400" counterweight spacing to move the center of mass
- Rebuilt the chassis flat for a lower center of mass; improved from early elimination to second place at Rice

Rice Eclipse – *Recovery Team Engineer* – Rice University

Sep 2025 – Feb 2026

- Collaborated on dual-stage test rockets exceeding Mach 2, focusing on safe and reliable recovery
- Designed recovery modules for airframe separation, pressurization, parachute deployment, and avionics mounting
- Applied OpenRocket, SolidWorks, MATLAB, and fabrication tools to iterate designs and produce launch-ready hardware